SENTINEL

An intrusion detection system that knows when to ask a human.

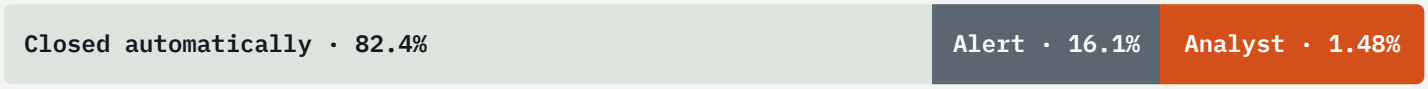
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Guide: **Prof. Divya Nimbalkar**

Most AI detectors decide every case alone and stay silent when wrong. Sentinel handles what it is sure about, asks a security analyst about the rest, and learns from verified decisions under rules fixed in advance.

WHAT HAPPENED TO 257,222 UNSEEN TEST FLOWS

held-out CIC-IDS2017 traffic, never used in training



212,015 sure-normal flows closed · **41,401** confident attack alerts · **3,806** doubtful flows for a human

98.9%
of the model's mistakes land in the analyst's queue

99.999%
correct on the traffic handled without a human

2
of 43,706 attack flows closed as normal

0
false alarms on 213,516 normal flows

HOW IT WORKS

01 A flow arrives
64 measurements of one conversation. No IPs or ports.

02 CNN-LSTM classifies
Normal, or one of 12 attack types.

03 Three checks
Hesitant? Unfamiliar? Second model disagrees?

04 Route it
Sure: closed or alerted. Any doubt: analyst.

05 Analyst decides
Verdict with reason, beside similar past cases.

The learning loop. An administrator verifies analyst decisions, retrains a candidate model, and deploys it only if it passes three rules set before training. Any deployment can be rolled back in one click.

DATA AND MODEL

Improved CIC-IDS2017, five days of labelled lab traffic: 2,099,976 flows, **1,714,812** after removing duplicates.



`Conv1D 64 → Pool → Conv1D 128 → LSTM 128 → 13 classes · + Random Forest (RF) 2nd opinion`

ACCURACY · MEAN OF 3 TRAINING RUNS

Attack vs normal	Paper	RF	CNN-LSTM
Detection rate	96.6%	99.92%	99.75%
Precision	92.1%	99.90%	99.71%
F1 score	94.3%	99.91%	99.73%
Macro-F1 (11 classes)	–	99.23%	98.09%

Indicative only: our data cleaning differs from the paper's. Trees stay slightly more accurate.

ATTACKS THE MODEL NEVER SAW IN TRAINING

share of the attack's flows closed as normal · lower is better

Web attack 97.0% → **5.0%**

Port scan 39.7% → **4.3%**

Botnet 12.4% → **10.2%**

Grey: CNN-LSTM alone. Orange: with Sentinel's routing. Brute-force SSH logins still pass: each login looks normal alone.

SECURITY ANALYST
Reviews doubtful flows with the reasons, unusual measurements and similar past cases, then records a verdict.

ADMINISTRATOR
Replays traffic, verifies feedback, retrains, deploys or rolls back models, and reads the audit log.

Python · PyTorch · scikit-learn · FastAPI · SQLite · React

Reference: de Nascimento & Hou, *Safety* 11(4):120, 2025